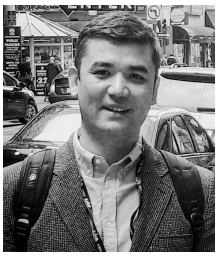
Berdakh Abibullaev



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berdakh - 0000-0002-8623-5526 - Berdakh Abibullaev
YouTube Channel

Quick Links:

- Short Biography
- Education
- Work Experience
- Peer Reviewed Publications
- Grants and Funding
- Honors and Awards
- Technical Skills
- Service to Scientific Community

Short Biography

Dr. Berdakh Abibullaev received his M.Sc. and Ph.D. degrees in electronic engineering from Yeungnam University, South Korea in 2006 and 2010 under the Korean Government Scholarship program. He held research scientist positions at Daegu-Gyeongbuk Institute of Science and Technology (2010-2013) and Samsung Medical Center, South Korea (2013-2014). He was also appointed as a research professor at Sungkyunkwan University, Seoul. In 2014, he received NIH (National Institute of Health, USA), a postdoctoral research fellowship to join a research project between the University of Houston Brain-Machine Interface Systems Team and a Texas Medical Center in developing neural interfaces for rehabilitation in post-stroke patients. Currently, he is an Assistant Professor in the Department of Robotics and Mechatronics, Nazarbayev University, Kazakhstan.

Dr. Abibullaev has considerable research and clinical experience working with patient populations in applying scientific and technical skills to advance the development of treatments for neurological disorders. His current research focuses on developing signal processing and machine learning models for the inference problems of Brain-Computer Interfaces, and brain data analysis.

Education

`2006-2010` **Ph.D., Electronic and Electrical Engineering**, *Yeungnam University*, Daegu, South Korea.

`2004-2016` **M.Sc., Electronic and Electrical Engineering**, *Yeungnam University*, Daegu, South Korea.

`2000-2004` **B.S., Information Technology**, *Tashkent University of Information Technologies*, Uzbekistan.

Experience

`09/2015-present` **Assistant Professor**, *Robotics & Mechatronics Department, School of Engineering and Digital Sciences, Nazarbayev University*, Nur-Sultan, Kazakhstan.

05/2018 - 07/2018 **Visiting Professor**, *Department of Electrical Engineering & Computer Science, University of Houston*, Houston, Texas, USA.

05/2014 - 09/2015 **NIH Postdoctoral Research Fellow II**, *Department of Electrical Engineering & Computer Science, University of Houston*,Houston, Texas, USA.

01/2014 - 05/2014 **Research Scientist**, *Department of Neurology, Samsung Medical Center, Sungkyunkwan University,*, Seoul, South Korea.

05/2011 - 12/2013 **Senior Research Scientist**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

02/2010 - 05/2011 **Postdoctoral fellow**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

Publications

*Publication list also available in Google Scholar.

- **Journal Publications**
- **Conference Publications**
- **Patents**

Journal Publications

B. Abibullaev and A. Zollanvari. **A Systematic Deep Learning Model Selection for P300-Based Brain-Computer Interfaces**, *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, DOI: 10.1109/TSMC.2021.3051136, 2021. [Link]

B. Abibullaev, I. Dolzhikova and A. Zollanvari. **A Brute-force CNN Model Selection for Accurate Classification of Sensorimotor Rhythms in BCIs**, *IEEE Access*, DOI: 10.1109/ACCESS.2020.2997681, 2020. [Link]

A. Zollanvari, M. Abdirash, A. Dadlani and B. Abibullaev. **Asymptotically Bias-Corrected Regularized Linear Discriminant Analysis for Cost-Sensitive Binary Classification**, *IEEE Signal Processing Letters*, 2019. [Link]

B. Abibullaev and A. Zollanvari. **Learning Discriminative Spatiospectral Features of ERPs for Accurate Brain-Computer Interfaces**, *IEEE Journal of Biomedical and Health Informatics*, vol. 98, pp.1-12, 2019. [Link]

B. Abibullaev, A. Zollanvari, B. Saduanov, and T. Alizadeh. **Design and Optimization of a BCI-Driven Telepresence Robot Through Programming by Demonstration**, *IEEE Access*, 2019, vol. 7. [Link]

B Abibullaev, J An, SH Lee, JI Moon. **Design and Evaluation of Action Observation and Motor Imagery based BCIs using NIRS**, *Measurement*, vol. 98, pp. 250-261, 2017, Elsevier. [Link]

N.A. Bhagat, A. Venkatakrishnan, B. Abibullaev, E.J. Artz, N. Yozbatiran, A. Blank, J. French, C. Karmonik, R.G.Grossman, M.K O'Malley, G. Francisco, J.L. Contreras-Vidal. **Design and optimization of an EEG-based brain machine interface (BMI) to an upper-limb exoskeleton for stroke survivors**, *Front. Neurosci.*, vol. 10, March, 2016. [Link]

J.G. Cruz-Garza, Z.R. Hernandez, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **A novel experimental and analytical approach to the multimodal neural decoding of intent during social interaction in freely-behaving human infants**, *Journal of Visualized Experiments*, doi:10.3791/53406, October, 2015. [Link]

C.H. Park, J.H Seo, D. Kim, B. Abibullaev, H. Kwon, Y.H. Lee, M.Y. Kim, K. Kim, J.S. Kim, E.Y. Joo, S.B. Hong, (2015, Feb). **Source Imaging in Partial Epilepsy in Comparison with Presurgical Evaluation and Magnetoencephalography**, *Journal of Clinical Neurology*, 2015 Feb 17, 11:e12. [Link]

B. Abibullaev, J An, S.H. Jin, and J.I. Moon. **Classification of brain hemodynamic signals arising from visual action observation tasks for brain-computer interfaces: An fNIRS study**, *Measurement*, 2014. Elsevier. [Link]

B. Abibullaev, J An, S.H. Lee, S.H. Jin, and J.I. Moon. **Minimizing inter-subject variability in FNIRS based brain computer interfaces via multiple-kernel support vector learning**. *Medical Engineering Physics*, 2013. Elsevier. [Link]

B. Abibullaev and J. An. **Classification of frontal cortex hemodynamic response during cognitive tasks using wavelet transforms and machine learning algorithm**, *Medical Engineering Physics*, 34(10):1394–410, 2012. Elsevier. [Link]

B. Abibullaev and J. An. **Decision support algorithm for diagnosis of ADHD disorder using electroencephalograms**, *Journal of Medical Systems*, 36(4):2675–2688, 2011. Springer. [Link]

B. Abibullaev, J. An, and J.I. Moon. **Neural network classification of brain hemodynamic responses from four mental tasks**, *International Journal of Optomechatronics*, 5(4):340–359, 2011. Taylor & Francis. [Link]

B. Abibullaev and H.D. Seo. **A new QRS detection method using wavelets and artificial neural networks**. *Journal of Medical Systems*, 35(4):683–691, 2011. Springer. [Link]

B. Abibullaev, M.S. Kim, and H.D. Seo. **Epileptic spike detection using continuous wavelet transforms and artificial neural networks**, 8(1):33–48, 2010. *International journal of wavelets, multiresolution and information processing*, Worldscientific. [Link]

B. Abibullaev, M.S. Kim, and H.D. Seo. **Seizure detection in temporal lobe epileptic EEGs using the best basis wavelet functions**. *Journal of Medical Systems*, 34(4):755–765, 2010. Springer. [Link]

M.S. Kim, Y.C. Cho, B. Abibullaev, and H.D. Seo. **Analysis of brain functions and classification of sleep stage EEG using daubechies wavelet**. *Sensors and Materials* , 20(1):1--15, 2008. MYU, Japan. [Link]

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Conference Publications

A. Oleinikov, B. Abibullaev, M. Folgheraiter, "**On the Classification of Electromyography Signals to Control a Four Degree-Of-Freedom Prosthetic Device**," in 2020 42nd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), July 20-24, Montreal, Canada.

B. Saduanov, D. Tokmurzina, K. Kunanbayev and B. Abibullaev, "**Design and Optimization of a Real-Time Asynchronous BCI Control Strategy for Robotic Manipulator Assistance**", in 2020 8th International Winter Conference on Brain-Computer Interface (BCI), 2020, GangWon, South Korea.

A. Tuleuov and B. Abibullaev, "**Deep Learning Models for Subject-Independent ERP-based Brain-Computer Interfaces**," in *the 2019 9th International IEEE/EMBS Conference on Neural Engineering (NER)*, March 20-23, 2019, San Francisco, CA, USA.

B. Abibullaev, Y. Orazayev, and A. Zollanvari, "**Novel Spatiospectral Features of ERPs Enhances Brain-Computer Interfaces** ," in *Brain-Computer Interface (BCI), 2019 7th International Conference IEEE, 2019, GangWon, South Korea.

B. Saduanov, D. Tokmurzina, T. Alizadeh, and B. Abibullaev, "**Brain-computer interface humanoid pre-trained for interaction with people**," in *2018 ACM/IEEE International Conference on Human-Robot Interaction*.1em plus 0.5em minus 0.4emACM, March 5-8, 2018, pp. 229–230, Chicago, IL, USA.

A. Oleinikov, B. Abibullaev, A. Shintemirov, and M. Folgheraiter, "**Feature extraction and real-time recognition of hand motion intentions from EMGs via artificial neural networks**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*.1em plus 0.5em minus 0.4em IEEE, 2018, pp. 1-5, GangWon, South Korea.

G. Lee, S. H. Jin, S. T. Yang, J. An and B. Abibullaev, "**Cross-correlation between HbO and HbR as an effective feature of motion artifact in fNIRS signal**," 2018 6th International Conference on Brain-Computer Interface (BCI), pp. 1-3, 2018, IEEE, GangWon, South Korea.

B. Saduanov, T. Alizadeh, J. An, and B. Abibullaev, "**Trained by demonstration humanoid robot controlled via a BCI system for telepresence**," in *Brain-Computer Interface (BCI), 2018 6th International Conference on*. IEEE, 2018, pp. 1–4, January, GangWon, South Korea.

G. Lee, S. H. Jin, S. H. Lee, B. Abibullaev, and J. An, "**fNIRS motion artifact correction for overground walking using entropy based unbalanced optode decision and wavelet regression neural network**," in *Multisensor Fusion and Integration for Intelligent Systems (MFI), 2017 IEEE International Conference on*.IEEE, 2017, pp. 186–193, Daegu, South Korea.

A. Zhumadilova, D. Tokmurzina, A. Kuderbekov and and B. Abibullaev. **Design and Evaluation of a P300 Visual Brain-Computer Interface Speller in Cyrillic Characters**. In *the 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN), IEEE*, 2017, Aug 28 - Sept 1, Lisbon, Portugal.

B. Abibullaev.**Learning Suite of Kernel Feature Spaces Enhances SMR-Based EEG-BCI Classification**. In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, GangWon, South Korea.

D. Nurseitov, A. Serekov, A. Shintemirov and B. Abibullaev. **Design and Evaluation of a P300-ERP based BCI System for Real-Time Control of a Mobile Robot**. In *the 5th International Winter Conference on Brain-Computer Interface, IEEE*, 2017, January 9-11, Gangwon , South Korea.

B. Abibullaev and J An. **On Robust Classification of Hemodynamic Signals for BCIs via Multiple Kernel *ν*-SVMs**. *Intelligent Robots and Systems (IROS), 2016 IEEE/RSJ International Conference*, 2016, Oct10-15, Daejeon, South Korea.

J.G. Cruz-Garza, Z.R. Hernandez, M. Megjhani, B. Abibullaev, T. Tse, E. Caducoy, and JL Contreras-Vidal.**Neural development of social cognition in the first two ears of life - Early findings**. In *Society for Neuroscience* 2015 Oct 17-21; Chicago, USA.

Z.R. Hernandez, A.J. Arenas-Castellanos, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal, **Decoding Intent From Non-invasive EEG in Freely Behaving Infants** In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, USA.

A.J. Arenas-Castellanos, Z.R. Hernandez, J.G. Cruz-Garza, M. Megjhani, B. Abibullaev, Sri. R.P. Maddi, T. Tse, C. Armstrong, W. Long, J.L. Contreras-Vidal, **A developmental Analysis of Behaviors Related to the Mirror Neuron System in 6-24 Months Infants**, In *Ninth Biennial Meeting of the Cognitive Development Society* 2015 Oct 9-10; Ohio, USA.

Z.R. Hernandez, J.G. Cruz-Garza, T. Tse, E. Caducoy, B. Abibullaev, J.L. Contreras-Vidal. **Supervised Classification of Intended Behaviors Using Electroencephalography (EEG) from Freely-Behaving Infants: Early Findings**. In *12th Annual GCC Conference on Theoretical and Computational Neuroscience*, Rice University, Houston, TX, United States, February 6-7, 2015.

N. A. Bhagat, A Venkatakrishnan, B. Abibullaev, E. J. Artz, A.A. Blank, J. A. French, N. Yozbatiran, and R. G. Grossman, M. K. OMalley, J. L. Contreras-Vidal, G. E. Francisco, **Control of a Therapeutic Exoskeleton to Facilitate Personalized Robotic Rehabilitation of the Upper Limb**. , Westin Arlington, Nov. 19-20 2014, United States.

C.H.Park, B. Abibullaev, E.Y. Joo, S.C. Hong, and S.B. Hong. **The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis**. In *the 19th Korean Epilepsy Congress*, Seoul, Republic of Korea, June 12-14 2014.

S.H. Lee, J. An, G. Jang, S.H. Jin, B. Abibullaev, and J.I. Moon. **Neural activity during observation, imagery, and execution of eating: An fNIRS pilot study**. In *the 19th Annual Meeting of the Organization for Human Brain Mapping*, Seattle, WA, United States, June 16-20 2013.

J. An, S.H. Jin, S.H. Lee, G. Jang, B. Abibullaev, J. Ahn, H. Lee, and J.I. Moon. **Cortical activation pattern for grasping during observation, imagery, execution, FES, and observation-FES integrated BCI: An fNIRS pilot study**. In *the 35th Annual Int. Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Osaka, Japan, July 3-7 2013.

J. An, S.H. Lee, S.H. Jin, B. Abibullaev, G. Jang, J. Ahn, H. Lee, and J.I. Moon. **The beginning of neurohaptics: Controlling cognitive interaction via brain haptic interface**. In *the 2013 IEEE International Winter Workshop on Brain-Computer Interface*, Gangwon Province, Korea, Feb. 18-20 2013.

B. Abibullaev, J. An, J.I. Moon, S.H. Lee, and S.H. Jin. **A study on the BCI-Robot assisted stroke rehabilitation framework using brain hemodynamic signals**. In *the 9th Int Conference on Ubiquitous Robots and Ambient Intelligence*, IEEE proceedings, Daejon, Korea, November 26-29,2012.

B. Abibullaev, J. An, S.H.Lee, S.H. Jin, J.I Moon, H.J. Lee. **Decoding of Brain Hemodynamic Responses for Brain-Computer Interfaces via Ensemble Support Vector Learning**. In *the Human-Computer Interaction Conference*, Gangwon-Do, Korea,

January 11-13,2012.

B. Abibullaev, W.S. Kang, S.H. Lee, J. An, and H.D. Seo. **Near-infrared spectroscopy in the analysis of functional brain activity during cognitive tasks**. In *2010 IEEE Sensors Conference*, Hawaii, United States, November 1-4, 2010.

B. Abibullaev, W.S. Kang, S.H. Lee, and J. An. **Recognition of brain hemodynamic mental response for brain-computer interface**. In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

S.H. Lee, B. Abibullaev, W.S. Kang, and J. An. **Analysis of attention deficit hyperactivity disorder in EEG using wavelet transform and self organizing maps**. In *International Conference on Control Automation and Systems*, IEEE proceedings, Gyeonggi-do, Korea, October 27-30 2010.

W.S. Kang, B. Abibullaev, S.H. Lee, and J. An. **Path planning algorithm using the values clustered by K-means**. In *the 15th Int. Symposium on Artificial Life and Robotics*, Japan, February 4-6 2010.

B. Abibullaev, H.D. Seo, W.S. Kang, and J. An. **A wavelet-based method for detecting and localizing epileptic neural spikes in EEG signals**. In *the 2nd Int. Conf. on Interaction Sciences: Information Technology, Culture and Human. ACM*, Seoul, Korea, Nov. 24- 26 2009.

B. Abibullaev and H.D. Seo. **Epileptic seizures detection using continuous time wavelet based neural networks**. In *the 6th International Conference on Information Technology: New Generations, IEEE Computer Society*, Las Vegas,Nevada, United States, April 27-29 2009.

W.S. Kang, B. Abibullaev, S.H. Lee, and J. An. **A study on brain activation during playing a computer game using fNIRS**. In *the 32th conference of Korean Info. Proc. Soc.*, Seoul Korea, November 22-24,2009.

B. Abibullaev, H.D. Seo, and M.S. Kim.**Classification system of EEG during cognitive mental tasks**. In *Int. Conference on Engineering and ICT*, Melaka, Malaysia, November 27-29, 2007.

H.D. Seo and B. Abibullaev. **Analysis of EEG signals by the continuous wavelet transforms**. In *the 5th Int. Joint Conference on Global Academic Networking*, Vladivastok, Russia, June 7-9 2007.

Patents

J. An, S.H. Jin, S.H. Lee, J.I. Moon, B. Abibullaev, J.H. Ahn and G.H. Jang. **Rehabilitation Training System and Method.** : 9,081,890, Washington, DC: United States. Patent and Trademark Office, 2015 [Link].

J. An, S.H. Jin, S.H. Lee, J.I. Moon, B. Abibullaev, J.H. Ahn and G.H. Jang. **Self-directed Rehabilitation Training Method Combining Brain Signals and Functional Electrostimulation**. : 17077057, *Application number*: 14049302, 09-OCT-2013, United States, [Link].

J. An, S.H. Jin, S.H. Lee, B. Abibullaev, J.I. Moon , J.H. Ahn and G.H. Jang. **Combining Brain Signals and Functional Electrostimulation Self-Directed Rehabilitation Method**, Patent Registration Number: 101501524000, 05.03.2015. South Korea, [Link].

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Grants and funding

Current Funding which includes only amounts over 40000 USD

01/2021-12/2023 **Principal Investigator**, *Brain-Computer Interface to Exoskeleton System Design for Personalized Upper-Limb Rehabilitation after Stroke*, Nazarbayev University.
This project aims to accelerate the efficacy of robotic rehabilitation using a new neural interface system developed for upper-limb repair after stroke. It aims to facilitate expediate motor recovery from incomplete upper-limb paralysis, and to investigate in-depth understanding of cortical reorganization in the brain associated with the recovery

01/2021-12/2023 **Co-Investigator**, *Classifier Validity and Its Applications in Bioinformatics,
The project studies the measures of classifier validity and explores their applications in bioinformatics. Nazarbayev University.

01/2019-12/2021 **Co-Principal Investigator**, *Neural-assisted optimization of human learning with multimodal neuroimaging of EEG and NIRS*, Nazarbayev University.
Implementing and validate a neural-assisted, machine-learning-based intelligent tutoring system that uses information from physiological signals to adapt the learning schedule to a user's individual learning capacity

Honors and Awards

11/2017 **The best paper award**, *fNIRS motion artifact correction for overground walking using entropy-based unbalanced optode decision and wavelet regression neural network*,
in Fusion and Integration for Intelligent Systems (MFI), 2017 IEEE International Conference, South Korea.

06/2014 **The best poster award**, *The evaluation of accuracy and clinical usefulness of 3D EEG source localization analysis*, Korean Epilepsy Congress 2014.

12/2012 **The best Researcher of the Year Award**, *Robotics Research Division, Daegu Gyeongbuk Institute of Science and Technology*, South Korea.

09/2006 **Korean Government Scholarship**, *Recipient of competitive Korean Government Information Technology Scholarship for International Graduate Students for a Ph.D. program*, South Korea.

09/2004 **Brain Korea Scholarship**, *Recipient of competitive graduate scholarship for a M.Sc. program*, South Korea.

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Technical Skills

Software : Expert in the Python Language and libraries such as Cython; very good knowledge of C, C++. Experience with a variety of tools and languages, including bash, LATEX, HTML, Git, Lab-streaming layer, various database query languages etc

Data Science : Expert in NumPy, Scipy, Sklearn, PyTorch, Tensoflow, Keras, MNE, Azure DevOps, AWS Certified.

Data acquisition systems : Expert user of Nicolet EEG Amplifiers, Blackrock Microsystems (intracranial recording), Functional Near-Infrared Spectroscopy FOIRE3000, BIOPAC fNIR200 Optical Brain Imaging, BioSemi Active-Two EEG system, Brain Products EEG amp., BIOPAC MP100 EEG amp. And G.tec g.USBamp systems.

Productivity applications : LaTeX, BibTEX, PSTricks, Beamer

System Environments : Linux (Ubuntu, Pop!_OS - favorite distro), Windows

Language Skills

- English (Professional proficiency)
- Korean (Professional conversational skills)
- Russian/Kazakh

Academic Service

Associate Editor

- IEEE Access
- PeerJ Computer Science

Program Committee Member

International Workshop on Artificial Intelligence for Healthcare Applications at International Conference on Pattern Recognition 2020.

The 7th International Conference on New Trends in Information Science, Service Science and Data Mining, June 18-20, 2013, Jeju Island, South Korea.

The 8th International Conference on Information Processing, Management and Intelligent Information Technology, April 1-3, 2013, Seoul, South Korea.

The 16th North-East Asia Symposium on Nano, Information Technology and Reliability, Oct.24-26, 2011, South Macao.

Session Chair

The 6th International Conference on Brain-Computer Interface (BCI), IEEE, 2018, South Korea

IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, 2017, Daegu, South Korea.

The 6th Int. Conference on Networked Computing, May 11-13, 2010, Gyeongju, South Korea.

International Conference on Control Automation and Systems Oct.27-30, 2010, Seoul, South Korea.

Occasional Reviewer

- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Journal of Biomedical and Health Informatics.
- IEEE Transactions on Systems, Man, and Cybernetics: Systems
- Brain Research, Elsevier.
- Neuroimage, Elsevier.
- Medical Engineering and Physics, Elsevier.
- Medical & Biological Engineering & Computing, Springer.
- ... other journals

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